Michał Płonka
Power BI & Data Analytics Portfolio

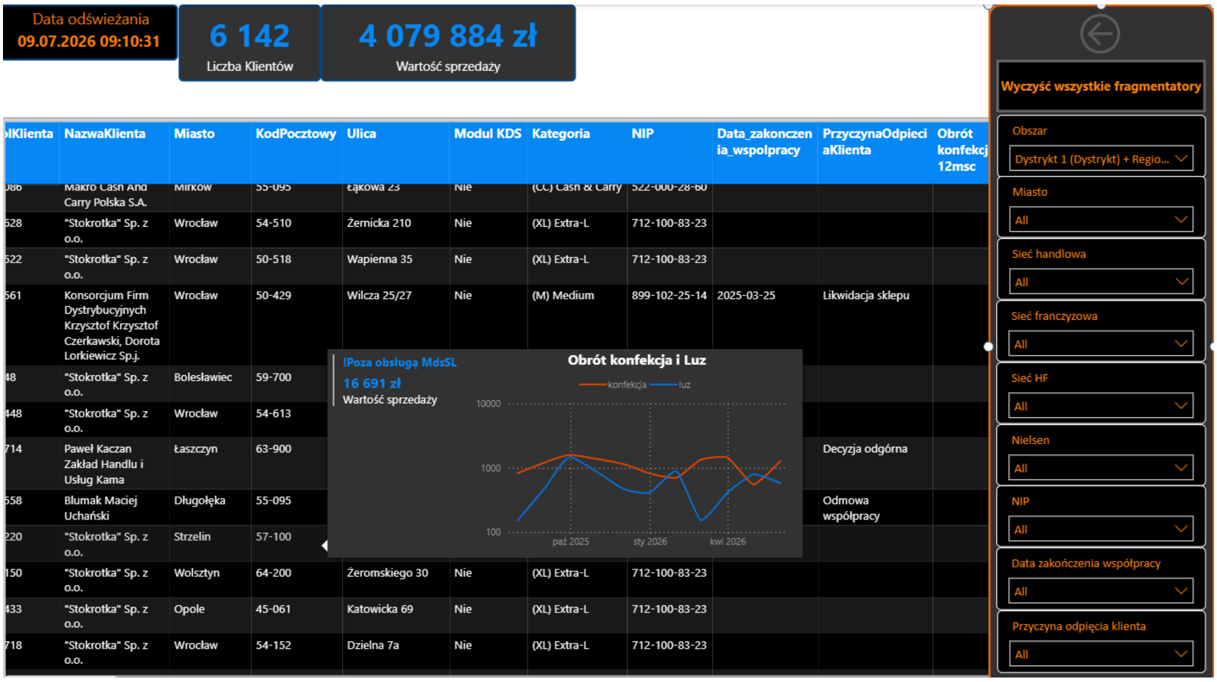
This portfolio presents selected Business Intelligence projects focused on data analysis, reporting automation, and decision support.
The reports were developed using Power BI and are based on structured data models, including star schema design and DAX calculations.
Each project demonstrates a combination of technical skills and business-oriented thinking.

Table of Contents

1 Potential Customer Reactivation & Coverage Dashboard

This Power BI report focuses on identifying potential customers for reactivation and field service prioritization, based on historical sales data and cooperation status.

The dashboard provides a comprehensive view of inactive or disengaged customers, enabling business teams to efficiently target high-value opportunities.



🔍 Key Features

KPI cards highlight the total number of customers and their aggregated sales potential.

These metrics allow quick estimation of the overall business impact.

A detailed table presents:

- customer identifiers (NIP, address, location),
- reason for churn / cooperation termination,
- date of last activity,
- historical turnover split (e.g., confection vs. bulk products).

This enables precise identification of reactivation candidates.

Customers are evaluated based on their 12-month historical turnover, allowing prioritization of high-value targets.

Advanced slicers allow filtering by:

- region / area,

- city,
- sales network,
- customer type,
- churn reason.

This supports both strategic and operational analysis.

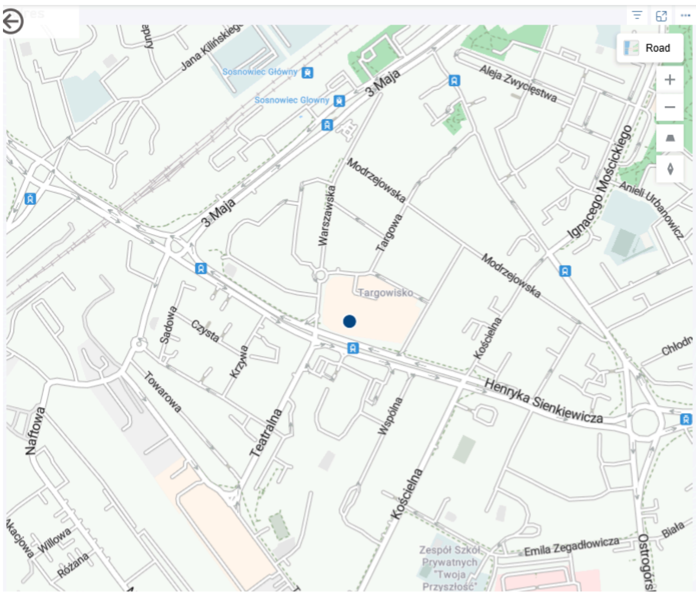
A time-based visual shows sales evolution across key product groups, helping detect behavioral changes before churn.

The report leverages drill-through functionality enhanced with geospatial analysis, allowing users to seamlessly transition from high-level summaries to detailed, location-based insights. Users can drill down and navigate to a map-based view, where the precise location of each customer is visualized.

The integration of drill-through with mapping significantly improves usability by:

- connecting analytical data with real-world locations
- reducing the need for external tools (e.g., Google Maps, CRM systems)
- supporting faster, data-driven operational decisions

👉 By combining drill-through navigation with geographic context, the report transforms raw data into actionable insights, particularly valuable for field teams and regional managers.



The report is built on a star schema data model, ensuring high performance, scalability, and clear semantic structure.

The model consists of a central fact table supported by multiple dimension tables, following best practices in analytical modeling. This structure:

- minimizes ambiguity,
- improves calculation performance,

- supports scalable reporting.

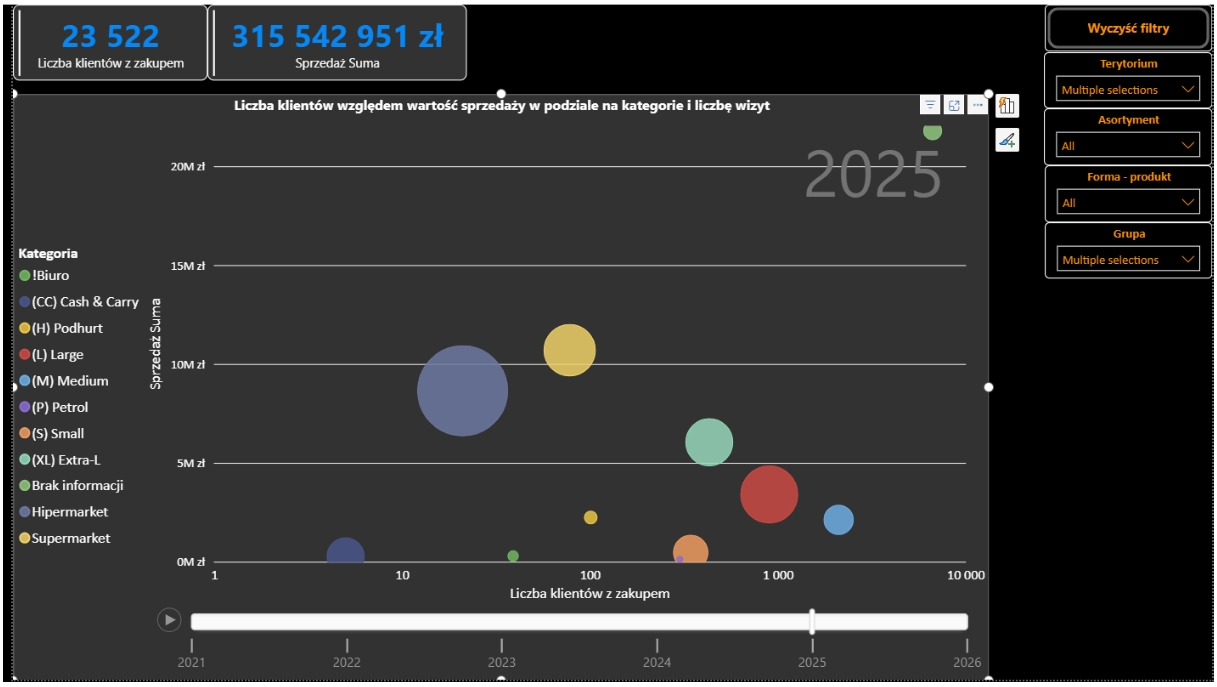
💡 **Business Value**

This dashboard supports:

- identification of high-potential customers for reactivation,
- prioritization of sales and field activities,
- recovery of lost revenue opportunities,
- better planning of customer outreach strategies,
- improved visibility into churn drivers.

2 Dynamic Customer & Sales Analysis (Rosling Chart)

This Power BI project features an interactive, animated bubble chart inspired by Hans Rosling's data storytelling techniques. It is designed to visualize the dynamic relationship between customer acquisition, sales revenue, and visit frequency across various business categories over time.



🔍 **Key Features**

Animated Time-Series Analysis: A dynamic playback axis (spanning from 2021 to 2026) allows users to observe the evolution of sales metrics, highlighting growth patterns and market shifts year over year.

Multi-dimensional Data Visualization: * X-Axis: Number of Purchasing Customers (Liczba klientów z zakupem) displayed on a logarithmic scale to properly distribute varying data magnitudes.

- Y-Axis: Total Sales Value (Sprzedaż Suma) in PLN.

- Bubble Size: Number of Visits (Liczba wizyt), serving as a clear indicator of customer engagement and traffic.
- Color Coding: Customer/Store Categories (e.g., Cash & Carry, Supermarkets, Petrol, Large, Medium), providing immediate visual segmentation.

Interactive Filtering: A dedicated slicer panel on the right enables deep-dive analysis by Territory, Assortment, Product Form, and Group. A custom "Clear Filters" button enhances the user experience.

Modern UI/UX: Designed with a sleek, professional dark theme that makes colors pop, featuring top-level KPI cards for quick metric overviews.

💡 Business Value

The dashboard enables stakeholders to quickly identify high-performing categories, spot underperforming segments, and understand the direct impact of foot traffic (visits) on overall revenue. By transforming static historical data into an engaging, animated narrative, it facilitates data-driven decision-making and trend discovery.

3 Data Quality Validator Dashboard

This Power BI report is designed as a comprehensive data validation and quality control tool, supporting analytical teams in ensuring the reliability and consistency of data across multiple sources.

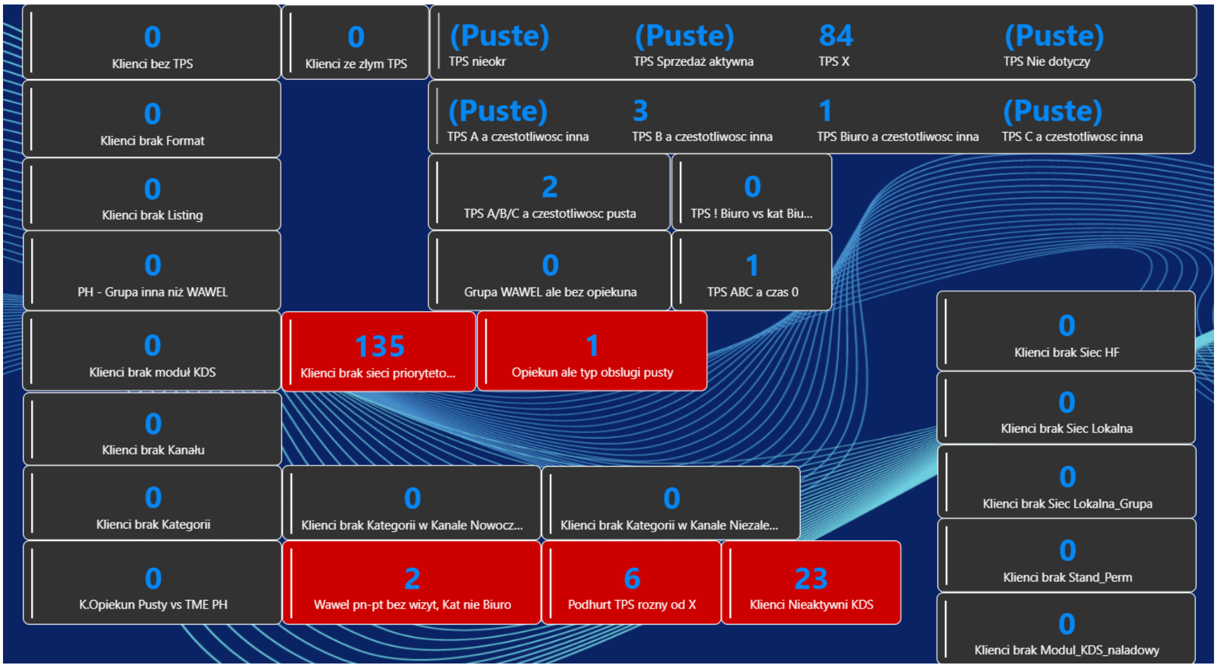
The solution integrates data from over 10 interconnected tables, forming a complex yet scalable model that enables systematic verification of data integrity across the organization.

🔍 Key Features

The primary goal of the report is to:

- identify data inconsistencies and anomalies,
- validate completeness and correctness of records,
- monitor data quality across multiple business domains,
- support ongoing data governance processes.

The dashboard allows users to quickly detect errors and investigate their root causes, significantly improving the overall trustworthiness of analytical outputs.



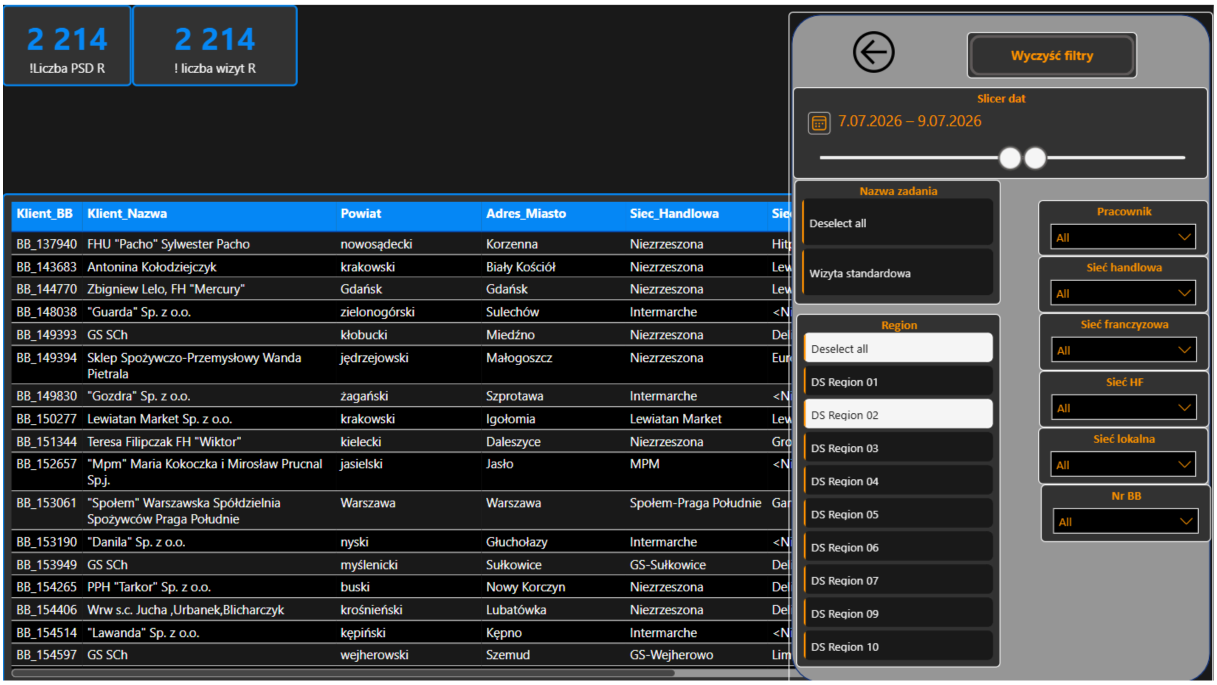
💡 **Business Value**

The Data Validator significantly:

- improves data reliability and report accuracy,
- reduces manual data checking efforts,
- accelerates error detection and resolution,
- enhances efficiency of BI and analytics teams,
- strengthens data governance practices.

4 Customer visits - planned and performed

This two-page analytical dashboard provides comprehensive tracking and optimization of field sales representatives' (Przedstawiciele Handlowi) activities. It bridges the gap between historical field execution and strategic future planning by splitting the analysis into two dedicated perspectives: Completed Visits and Planned Visits.



🔍 Key Features

Page 1: Historical Performance

This page provides a granular audit of completed field operations, allowing management to track field efficiency, representative performance, and time-on-site metrics.

- High-Level KPIs: Bounded by clear visual containers, the report instantly surfaces operational volume, tracking the total number of unique active points of sale (Liczba PSD) and the overall volume of field interactions (Liczba wizyt).
- Granular Activity Logging: The core matrix details exact operational data per visit, including client demographics (City, Street), assigned representative, exact start/end timestamps, and calculated Visit Duration (Czas trwania wizyty) to monitor workforce productivity.
- Advanced Multi-Layered Filtering Panel: Features a robust right-hand slice architecture enabling deep-dives by:
  - Task type (Nazwa zadania) and Region.
  - Individual Sales Representative (Pracownik).
  - Retail structures, allowing segmentation by Key Accounts (Sieć handlowa), Franchise networks (Sieć franczyzowa), and Local chains (Sieć lokalna).
- UX Optimizations: Includes a global "Clear Filters" (Wyczyść filtry) button and a precise dual-slider date range picker to dynamically isolate specific campaigns or fiscal quarters.

Page 2: Forward-Looking Planning & Pipeline Scheduling

- **Future Route & Schedule Visibility:** Displays upcoming, scheduled visits to ensure optimal geographic coverage and prevent gaps in client relationship management.
- **Resource Allocation:** Allows coordinators to evaluate sales rep workloads for the coming weeks, ensuring high-priority accounts receive adequate face-time.
- **Target vs. Actual Preparation:** Provides a clear baseline to compare upcoming plans against past performance, allowing teams to adjust frequencies based on historical visit duration data from Page 1.

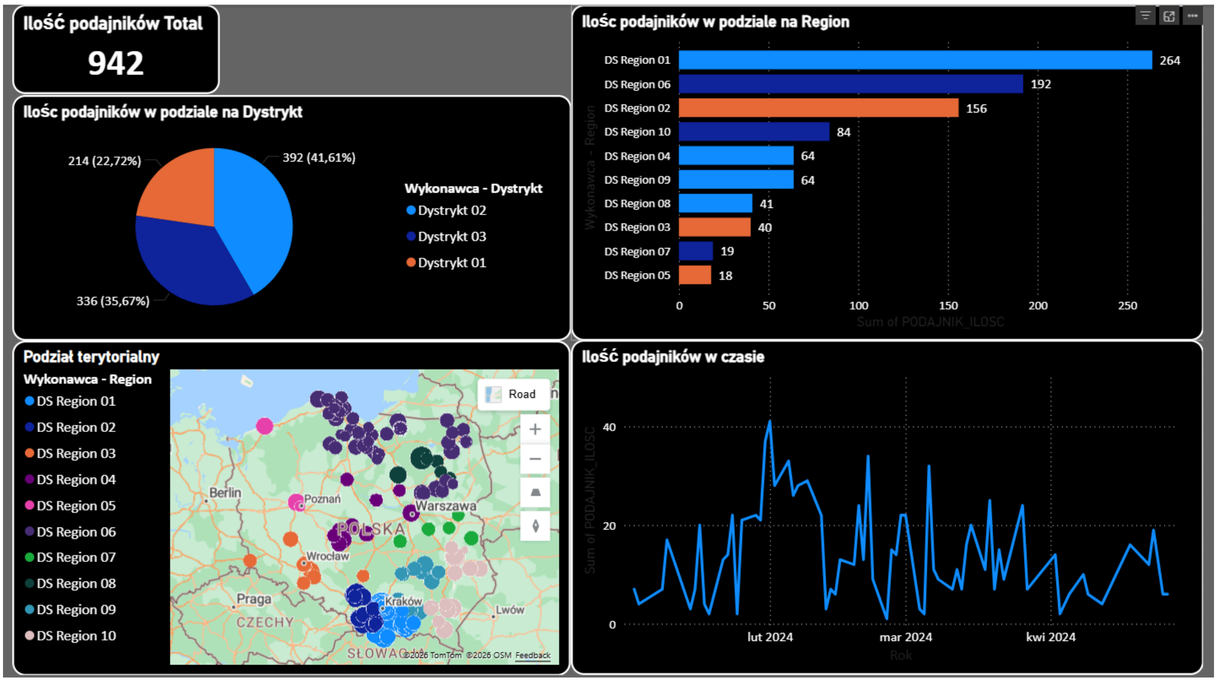
💡 Business Value

By transforming raw CRM and GPS-log data into actionable insights, this report significantly improves field force accountability, optimizes travel routes, reduces administrative overhead, and ensures that the retail execution strategy aligns perfectly with corporate targets.

5 Customer Feedback Survey Dashboard (chocolate wavelet)

This Power BI report presents the results of a chocolate wavelet-based customer survey, designed to monitor the performance and operational coverage across different organizational areas.

The dashboard provides a clear and interactive overview of key indicators, including total counts, regional distribution, and time-based trends.



🔍 Key Features

A pie chart shows the share of responses across districts (e.g., District 01, 02, 03), allowing quick comparison of contribution levels.

A bar chart highlights regional performance, with clear ranking (e.g., DS Region 01 leading with 264 entries).

A time-series chart tracks changes in activity over time making it possible to identify peaks, drops, and operational variability.

A top-level KPI card displays the total number of recorded entries (942), providing a quick snapshot of scale.

The report leverages drill-through capabilities, enabling users to move from high-level regional views into detailed data for each selected region.
This allows for:

- analyzing specific regions in depth,
- identifying local patterns and discrepancies,
- supporting operational decision-making at a granular level,
- link to photo.

💡 **Business Value**

This solution supports:

- performance monitoring across regions,
- identification of underperforming areas,
- data-driven decision-making,
- improved visibility of field operations over time.